

Mini-Maglev

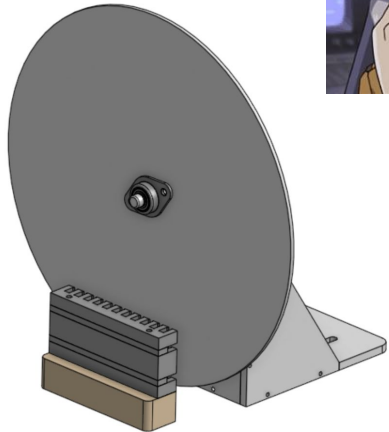
Capstone Proposal - ENPH 459

Team 2504: Ada Li, Amy Li, Cyrus Young, Dhruv Arun



Sponsor: ENPH Project Lab, UBC
Hennings Bldg, 6224 Agricultural Road

Dec 12, 2024



Agenda

1. Objective
2. Fundamentals
 - a. E&M
 - b. What are induction motors and how do they produce torque?
 - c. Why Field-Oriented Control?
3. Performance benchmarks, deliverables
4. Project timeline, risk analysis
5. Full Project Plan (system level diagram)

Appendix (not covered in this presentation)

6. Similar methods
7. Incremental test setups
8. Rough budget

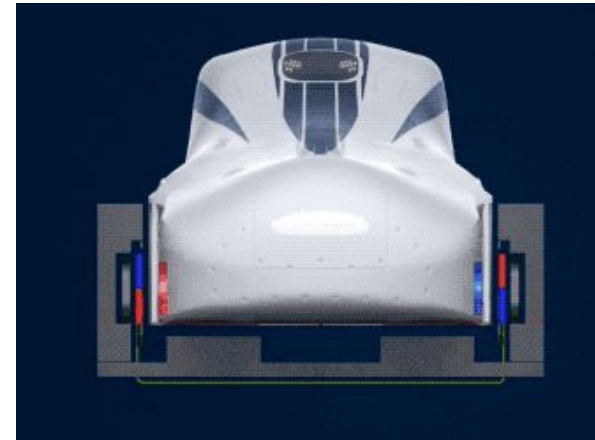
Introduction

High Level Overview

Technical goal: make a **mini model of a maglev train** using **linear induction motors**, and precisely control its speed!

- **Maglev:** train levitated above rails by magnetic repulsion
- **Induction motor:** torque-producing currents in rotor are induced by time-varying magnetic fields from stator
- **Linear:** the motor produces a linear force and motion directly on the train (which is the rotor), NOT torque and rotary motion of wheels.

Magnetic levitation is nice-to-have but not required, as long as we have linear motion (like Skytrain!)



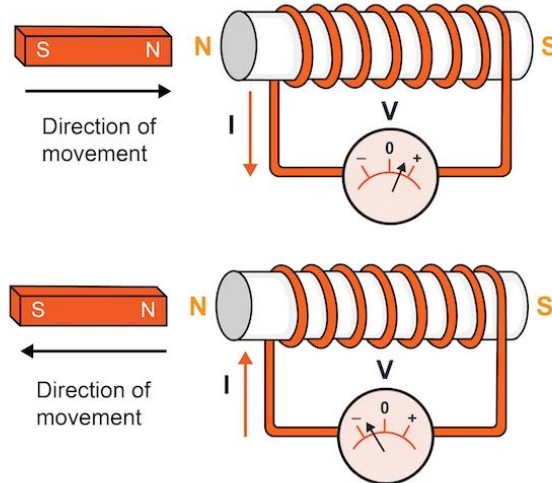
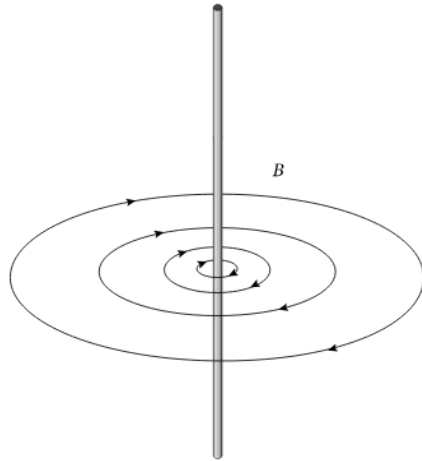
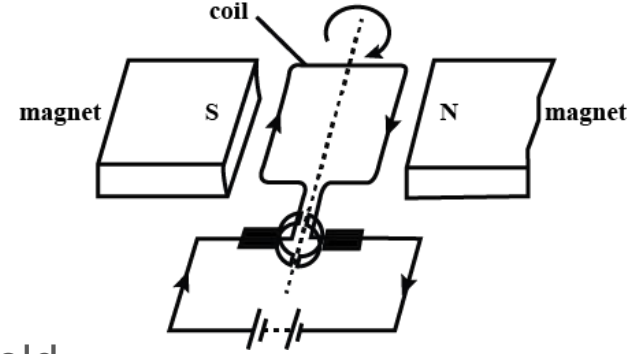
Sponsor and Project Motivation

- Sponsored by Engineering Physics Project Lab!
- Main purpose to provide us highly educational experience in practical design
 - NOT to produce a useful or commercializable product.
- Still want to produce a working prototype of good quality.

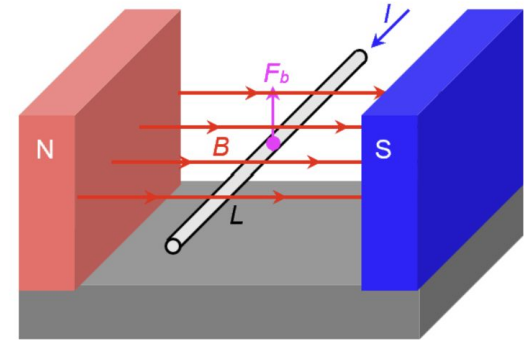
System Deep Dive

The Bare Basics

- Currents produce magnetic fields
- A changing magnetic field can induce current
- Moving charges experience a force in a magnetic field

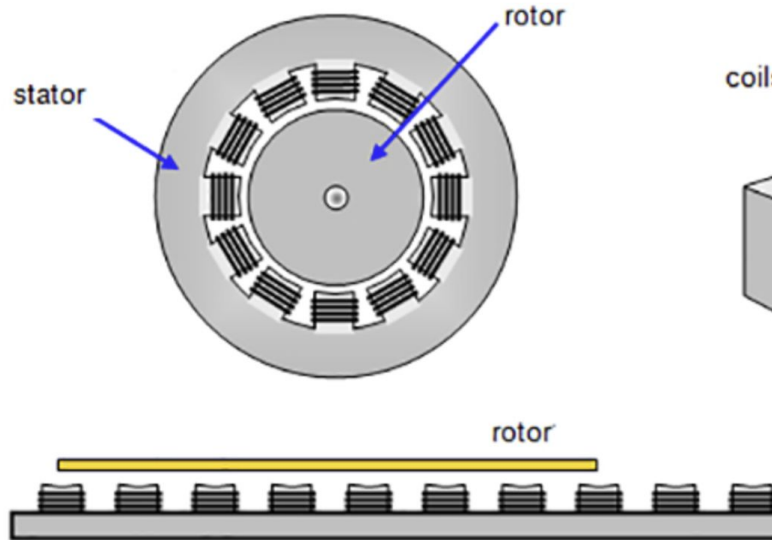


$$F_b = BIL$$

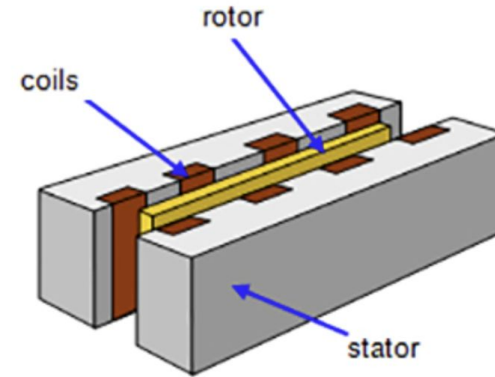


Induction Motors:

Regular Induction Motors:

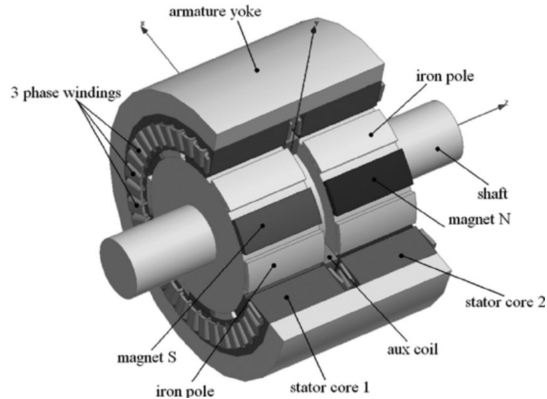


Linear Induction Motors:



Overview of Induction Motors - I

- Electric motor that uses AC to convert electrical energy into mechanical energy.
- Consists of Rotor and Stator:



https://www.researchgate.net/figure/ECPSM-with-the-surface-mounted-P-M-rotor-and-stator-structure-exhibiting-the-three-phase_fig1_256438728

<https://www.youtube.com/watch?v=TVQIszEyDo4>

- Changing B in stator produces a current in rotor by induction principle
- Rotor current pushes off magnetic field, producing torque

Overview of Induction Motors - II

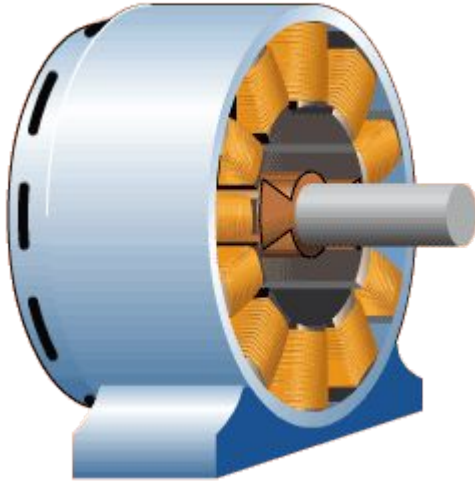
- AC (three-phase) supplied to stator → creates rotating B around stator windings.
- B rotates at synchronous speed which depends on AC supply frequency and number of magnetic poles in motor.

$$N_s = \frac{120f}{p}$$

- Rotating B induces current in rotor → Faraday's law of electromagnetic induction, i.e. a changing magnetic field induces an electric current in a conductor.

$$\varepsilon = -N \frac{\Delta \Phi}{\Delta t}$$

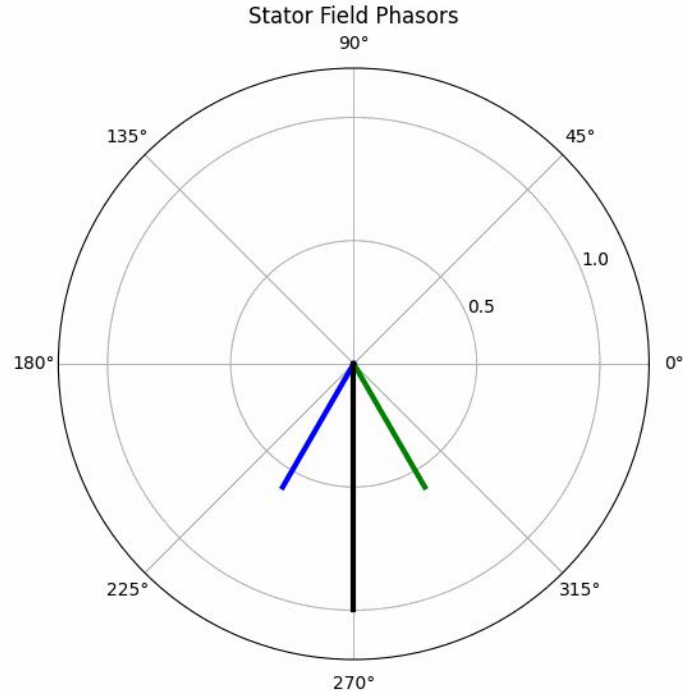
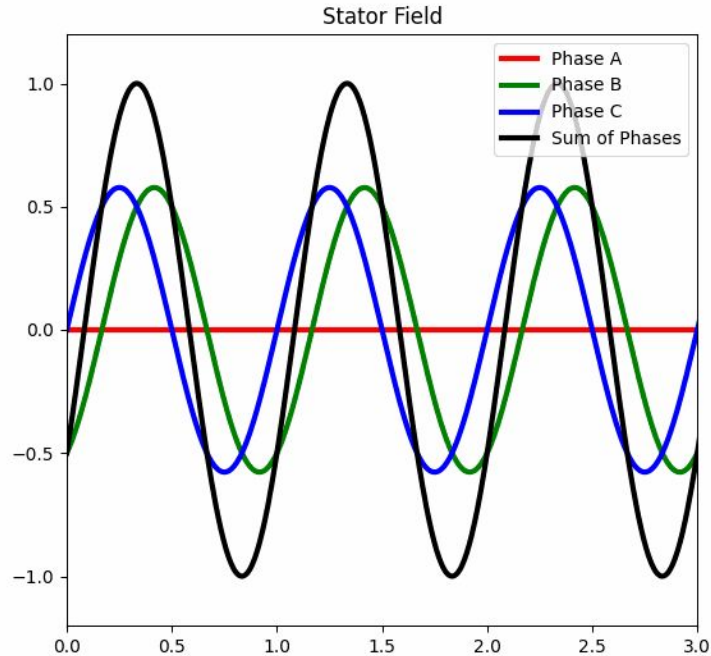
Simplified Induction Motor



<https://blog.opto22.com/optoblog/automation-101-adding-control-to-ac-motors>

Stator Travelling Flux Wave

- AC currents in the 3 phases create a travelling flux wave (in black)



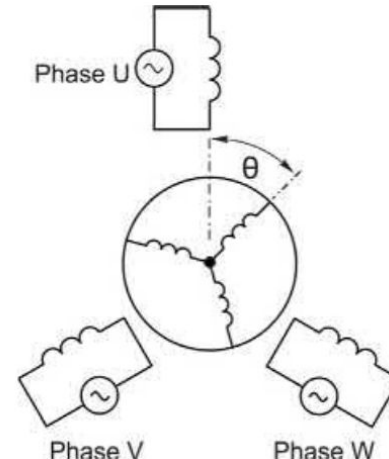
Overview of Induction Motors - III

- Current induced in rotor windings interacts with rotating B from stator, creating a force on rotor → rotor turns in direction of rotating B.
- Mutual flux → B generated by current in one coil passes through second coil
- Total torque requires nine terms (corresponding to each stator-rotor pair)

$$\Psi = n\Phi$$

$$M_{SR} = \frac{\text{Mutual flux linkage}}{\text{Current}} = \frac{\psi_{SR}}{i_S}$$

$$T = \sum i_S i_R \frac{dM_{SR}}{d\theta}$$

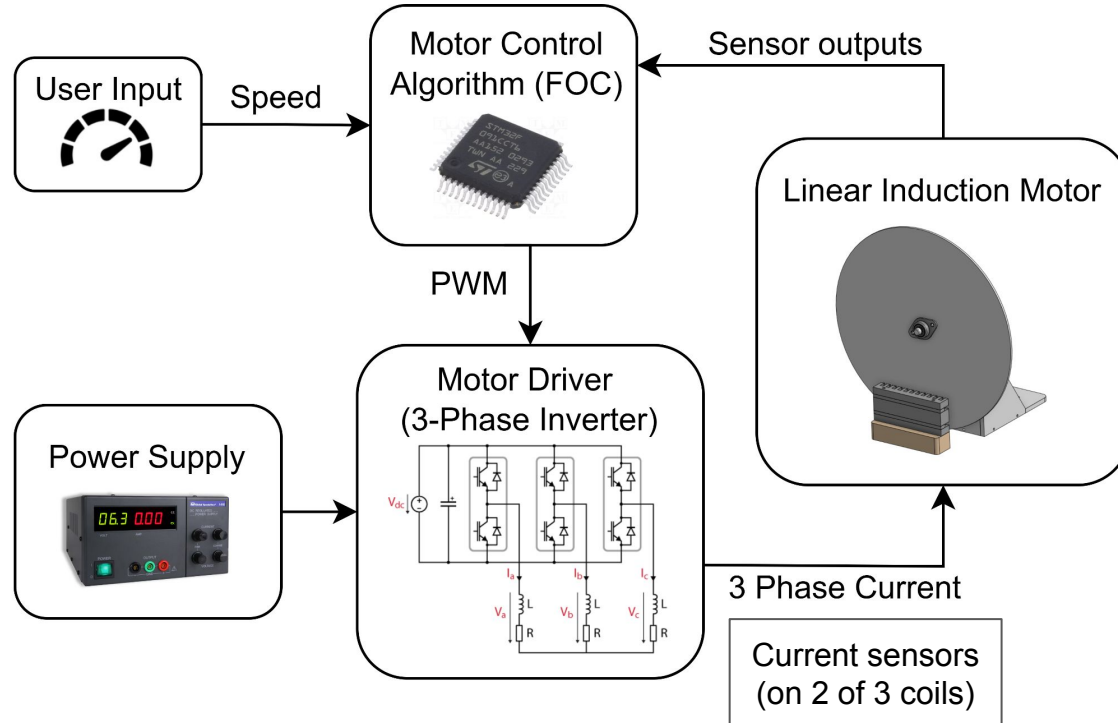


Overview of Induction Motors - IV

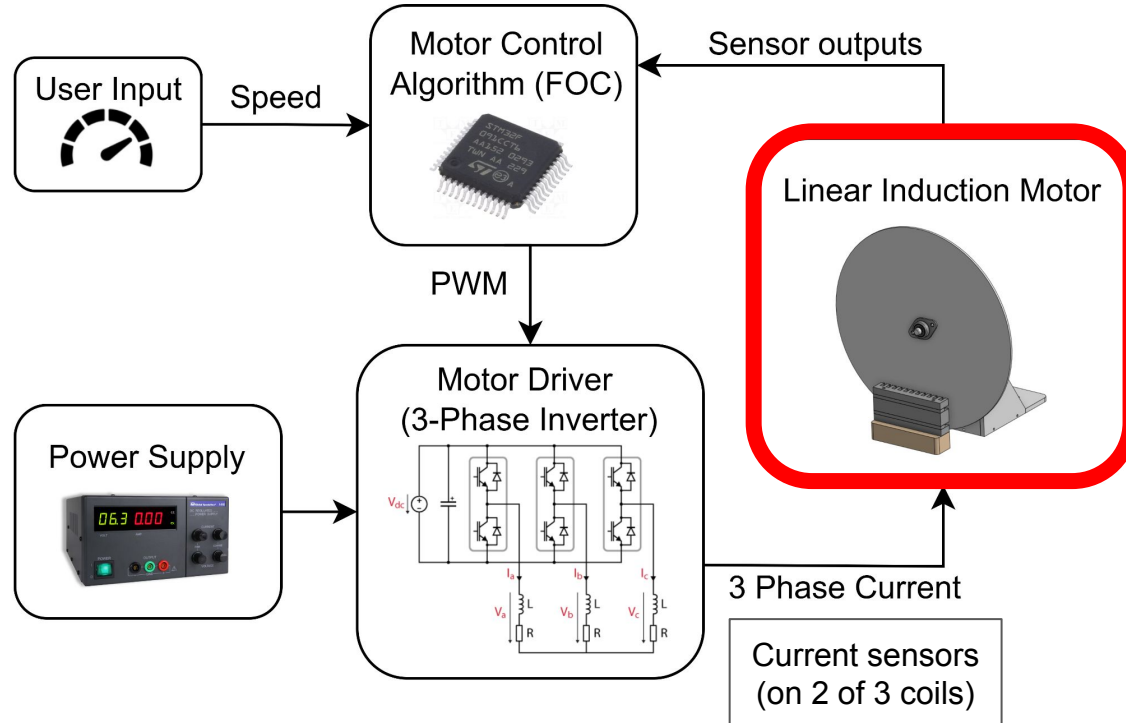
- Unlike synchronous motor, rotor in induction motor never catches up with synchronous speed of the stator's B
- Instead, it rotates at a slightly lower speed ("slip").
- Necessary for continuous induction and torque generation.
- Slip allows motor to generate relative motion between rotor and B of stator, ensuring ongoing induction and operation.

$$s = \frac{N_s - N}{N_s}$$

Simplified System Level Diagram



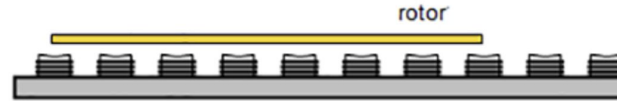
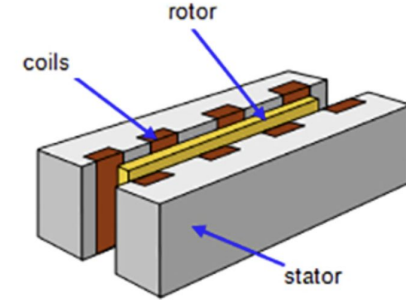
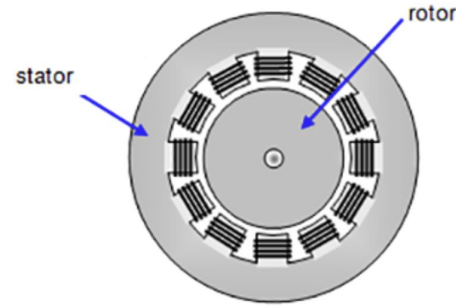
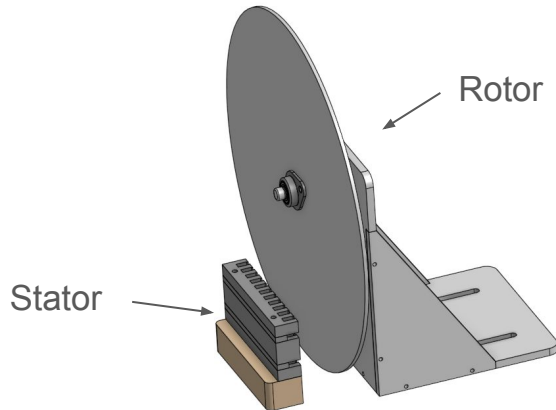
Simplified System Level Diagram



Overview of Linear Induction Motors - I

Consists of:

- Primary part (analogous to the stator in a rotary induction motor)
- Secondary part (analogous to the rotor in a rotary version).



<https://www.linearmotiontips.com/what-are-linear-induction-motors/>

Overview of Linear Induction Motors - II

- Coils effectively loop around the teeth of the track

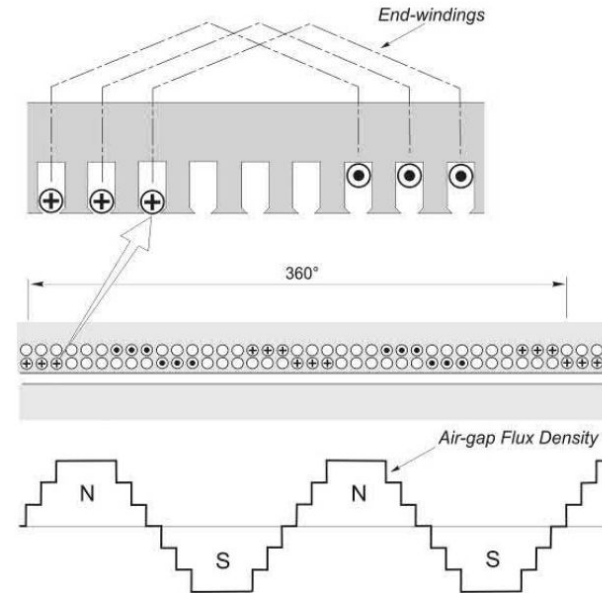
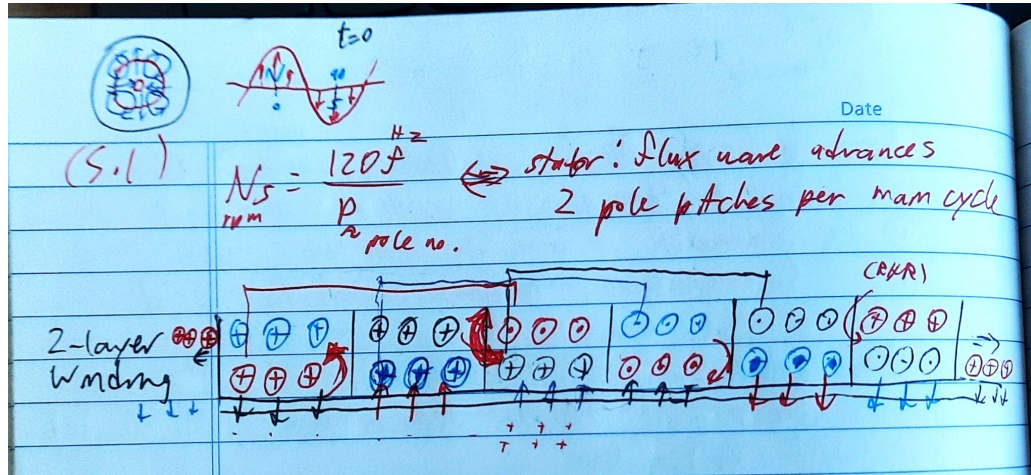
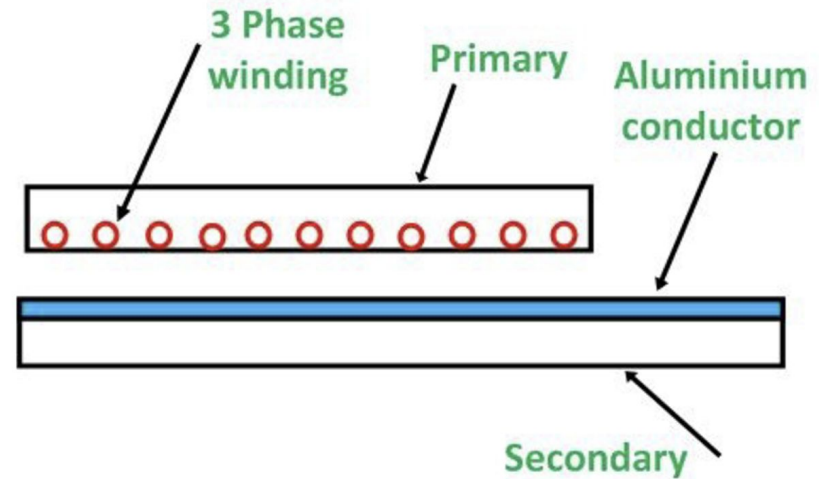


FIG. 5.5 Developed diagram showing layout of windings in a three-phase, four-pole, two-layer induction motor winding, together with the flux density wave produced by one phase acting alone. The upper detail shows how the coil-sides form upper and lower layers in the slots.

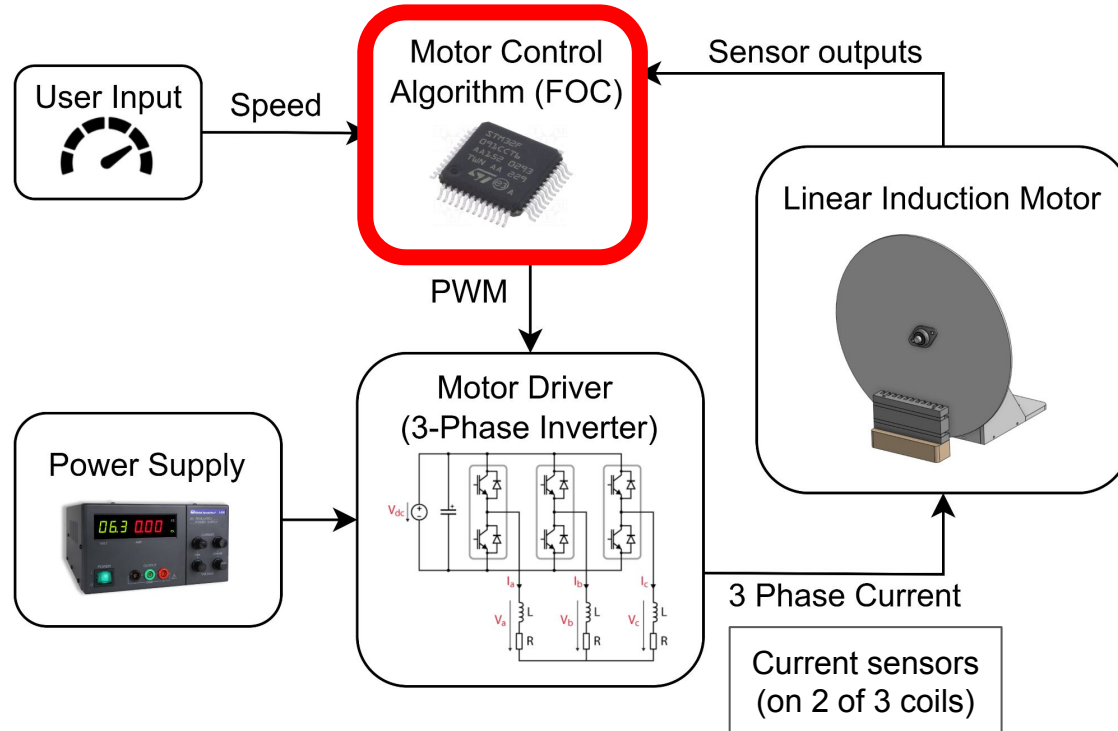
Overview of Linear Induction Motors - III

- Power supplied to primary part, Φ develops and travels across length of primary part.
- Eddy currents induced in the conducting material of the secondary part.
- Φ of primary part and induced currents of secondary part interact to produce a linear force



<https://www.linearmotiontips.com/what-are-linear-induction-motors/>

Simplified System Level Diagram



Field-Oriented Control Roadmap

1. What is FOC and why is it necessary?
2. The flux- and torque-producing components of the stator current.
3. How to control the stator current.

1. What is FOC?

- A VFD (Variable-Frequency-Drive) Control Method
- Manages the stator currents of a three-phase induction motor, where the AC signals are spaced 120 degrees apart in phase
- It represents stator currents as orthogonal components (explored further through Clarke and Park transformations)

Steady-State Speed Control by Varying Frequency

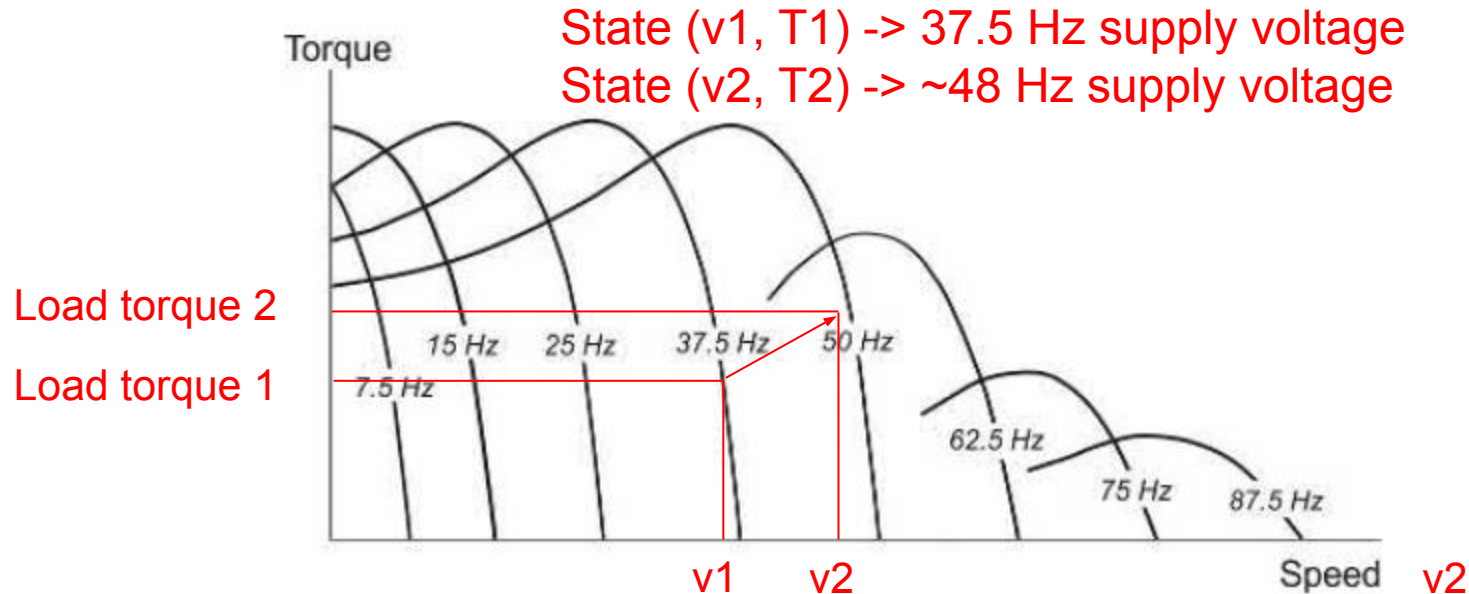
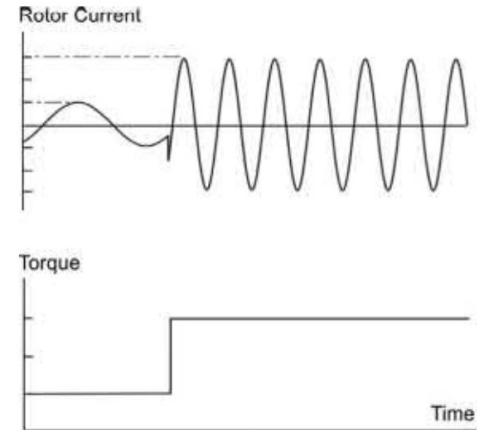
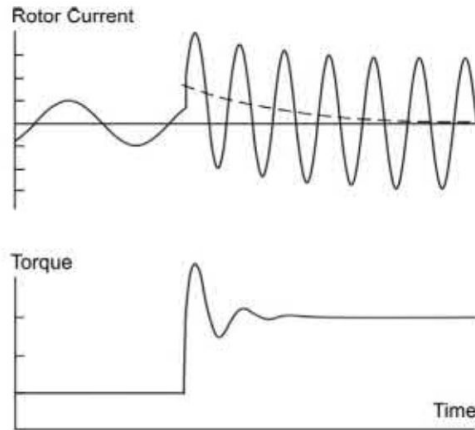


FIG. 7.4 Typical torque-speed curves for inverter-fed induction motor with constant flux up to base speed (50Hz) and constant voltage at higher frequencies.

FOC Advantage:

- The FOC advantage: (almost) instantaneously change the motor's torque output
 - This is done by changing the **frequency** and **phase** of the applied voltage such that the flux (i.e., energy) of the system does not change



2. Flux and Torque Components

- For FOC, 1st component defines the magnetic flux
- 2nd component describes the torque

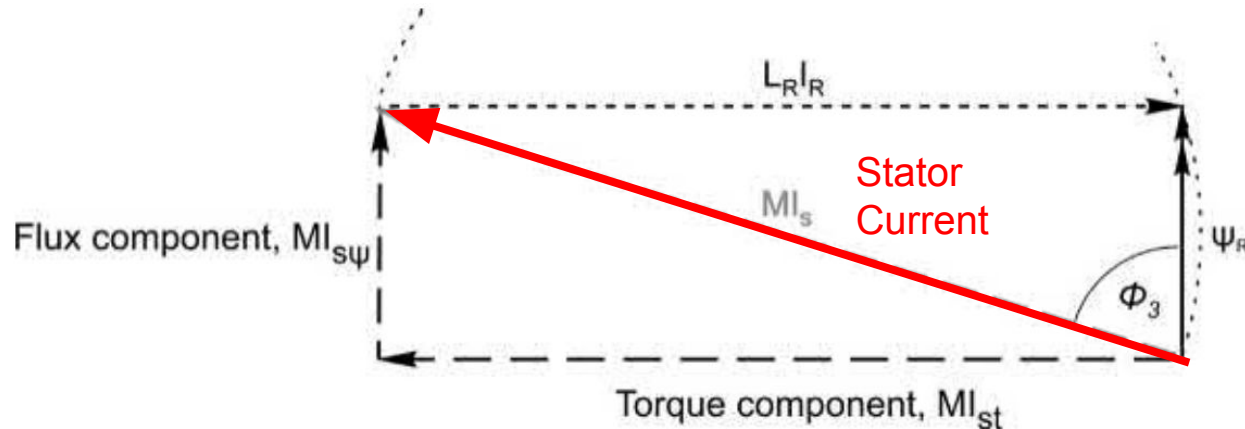
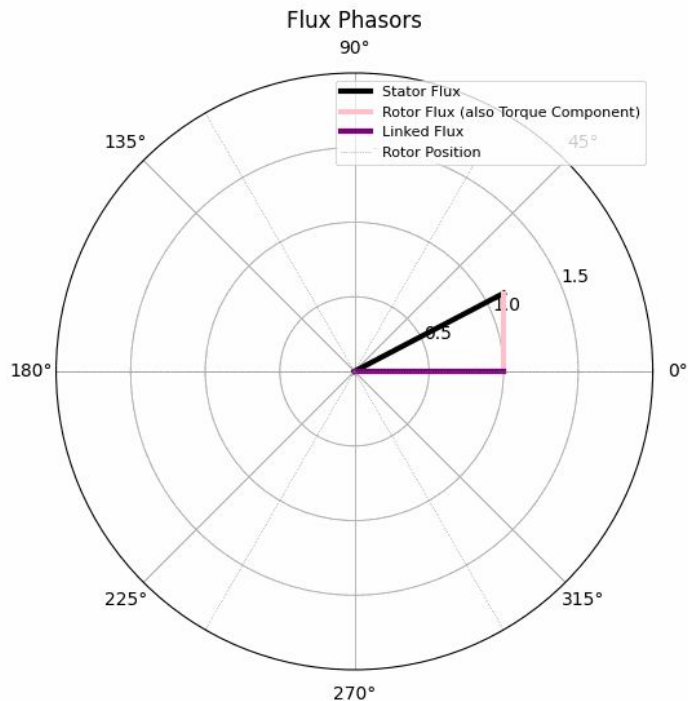


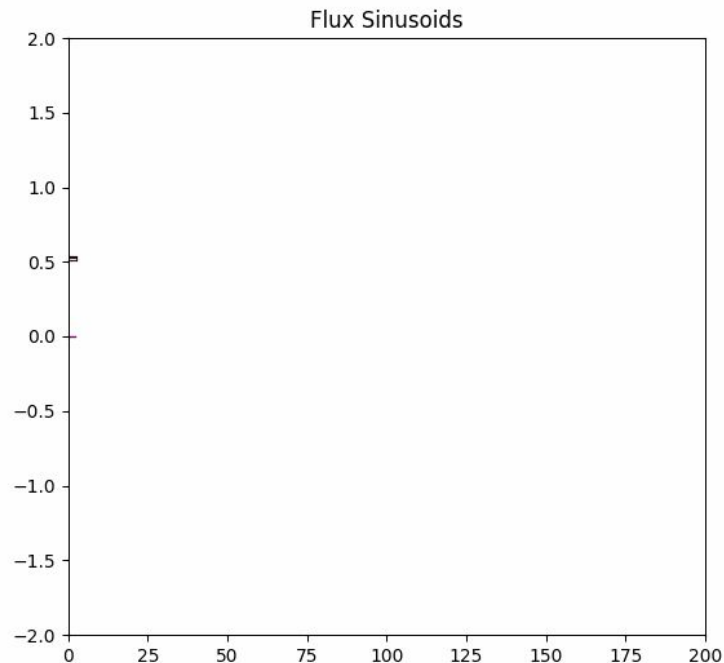
FIG. 8.9 Flux and Torque components of stator current.

3. How does FOC work?

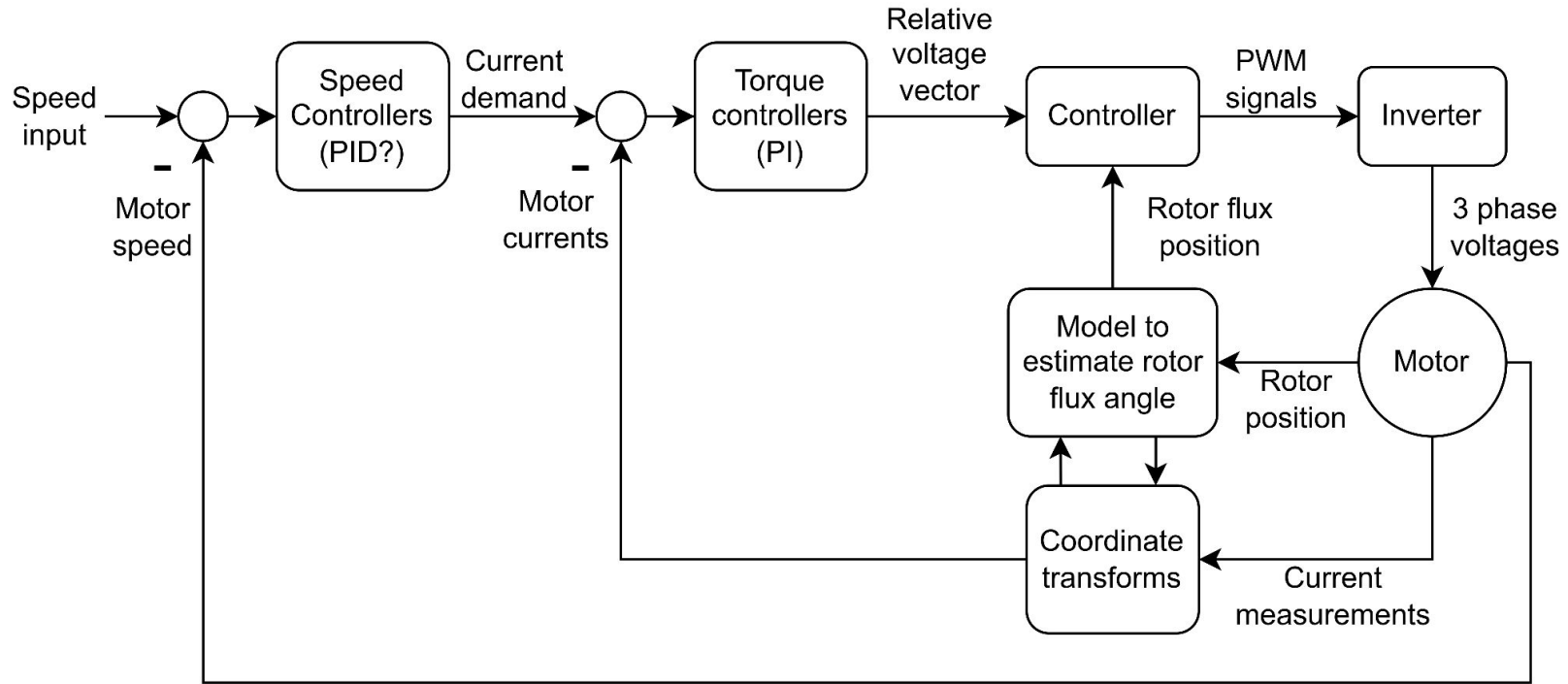
- Visualization of how FOC accomplishes instantaneous torque control



$$\tan \phi = \omega_s \tau$$



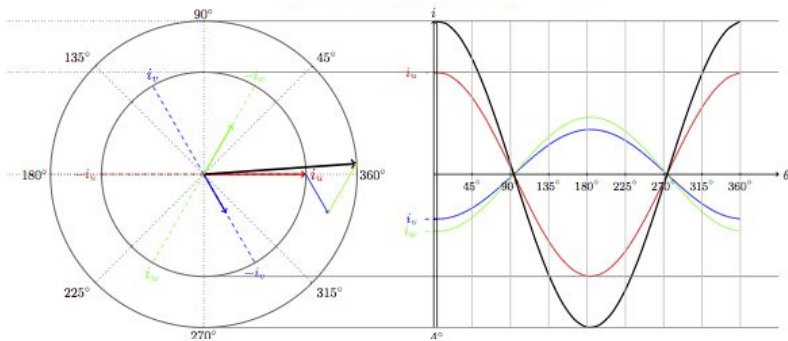
FOC Controller Details



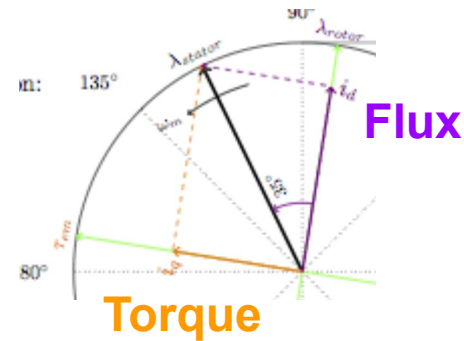
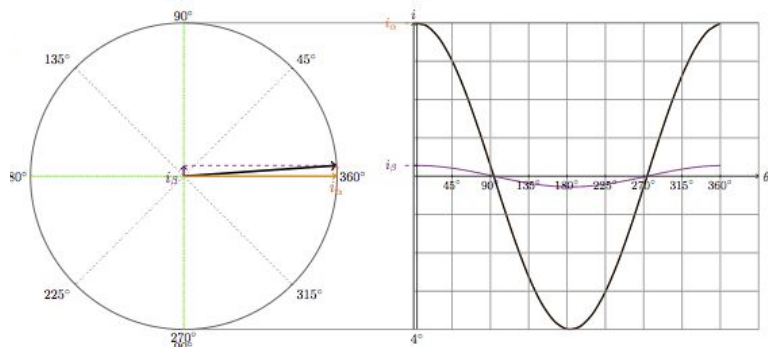
Coordinate Transforms

- Through the Clarke and Park transforms, we separate out the torque and flux components

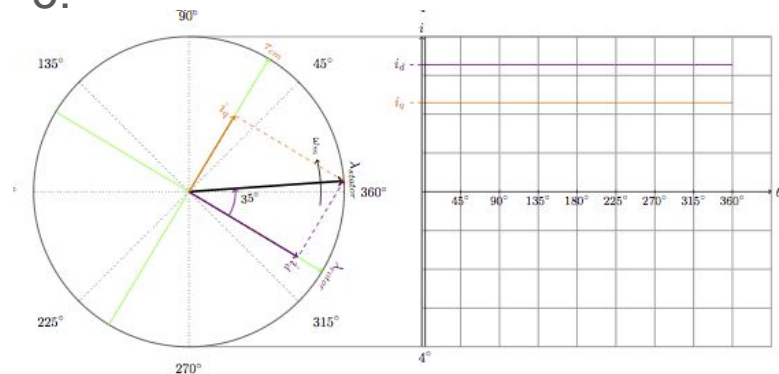
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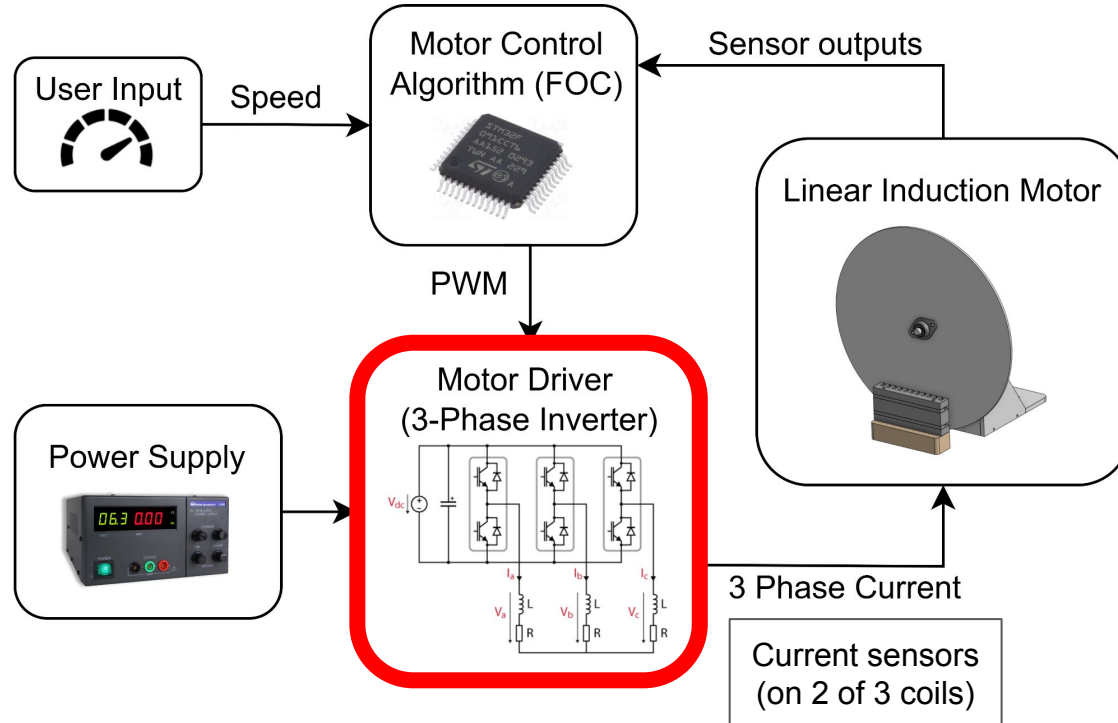
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3.

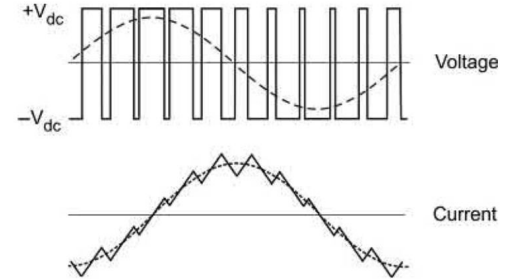


Simplified System Level Diagram

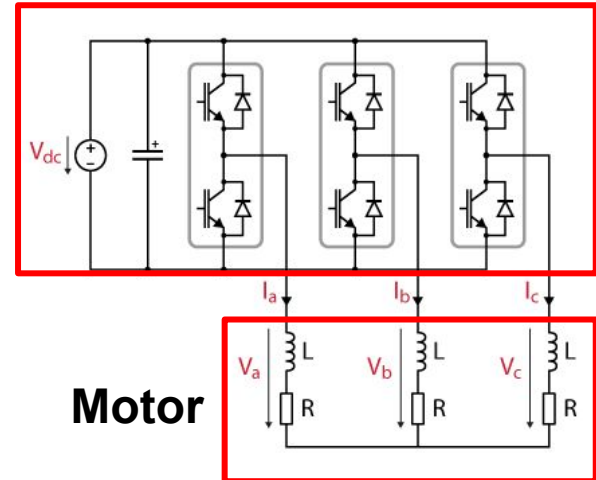


The Inverter

- Stator voltage is varied by driving PWM signals to a MOSFET gate driver:
- The gate driver drives the 3-phase inverter, which produces a 3-phase AC current with the correct frequency and phase.



3-phase inverter



Motor

What determines our Sample/control loop rate?

- Current control loop must run $\sim 10\times$ faster than the outer speed control loop, and **any electrical time constant** in the circuit.
 - Frequency limited by **code execution speed on CPU**
- Inverter control can run a bit slower than current control, but should still pull the latest voltage setpoint
 - Frequency limited by **switching losses** in inverter

Task (IDs 1-5)	Frequency	Priority
Update current setpoint vs speed error	500 Hz	0
Update voltage phasor setpoint vs current error & estimate of flux angle from current feedback	20 kHz	1
Switch inverter (PWM & vector modulation loop)	10 kHz	2
Encoder pulse?	External trigger	3
Update sensor?	External trigger	3

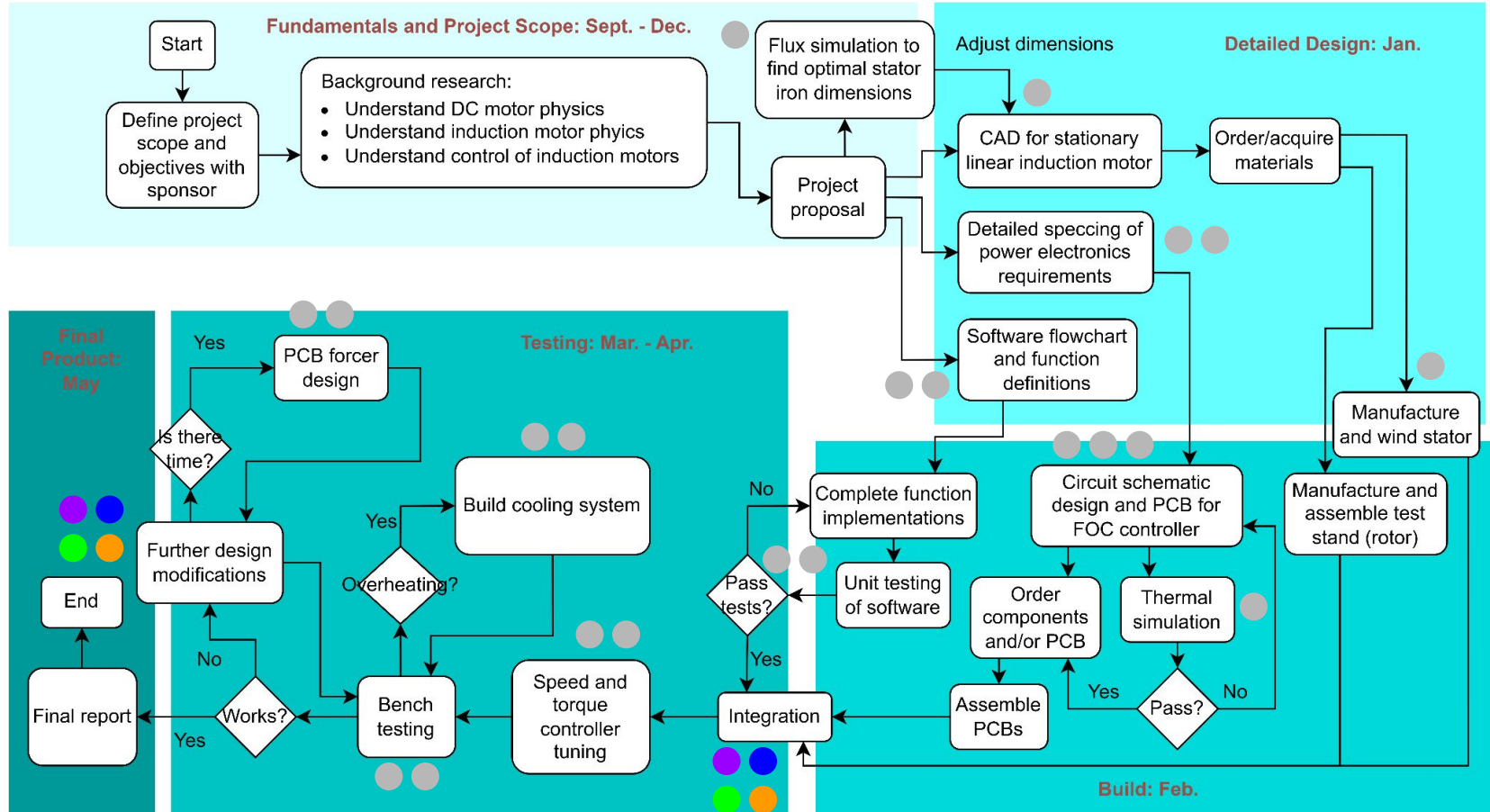
Performance Benchmarks and Deliverables

Performance Benchmarks

Target Benchmark	Evaluation Method
Torque control loop (i.e., sampling rate) running at 20(?) kHz	Measure control loop execution time in simulation or hooked up to dummy sensors (or verify that RTOS tasks are running as scheduled)
PWM switching (ideally > 16 kHz) occurs fast enough that current ripple is less than 2%	Rough calculations from motor time constant (pre-design) and hook up to oscilloscope (post-design)
Sensors (current, voltage, position) are accurate to 1% error	Calibrate and compare against known measurements
Flux estimation accurate to 1% error	Compare estimation against external flux sensor measurement
Sufficient heat sinking such that no component exceeds 80 degrees C	Test at max speed and torque, and monitor temperature sensors
Torque changes are transient-free	Monitor and plot stator currents
Can achieve ___ torque at ___ max speed	Dynamometer measurements

Project Timeline Flowchart

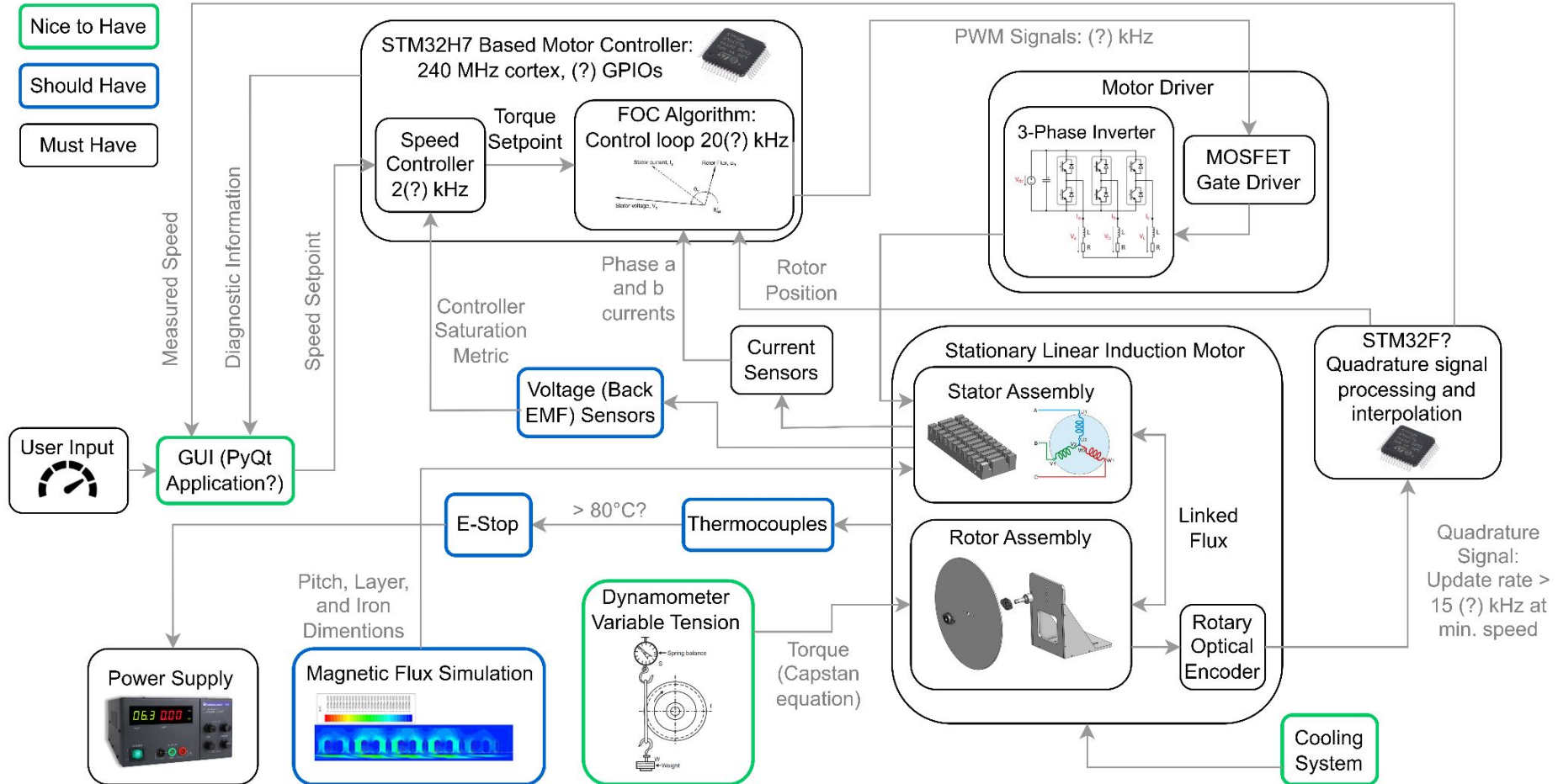
Ada ● Amy ● Cyrus ● Dhruv ●



Risk Analysis

Risk	Likelihood	Impact	Ability to mitigate	Score	Comments
Motor overheats and melts	4	7	9	3.1	Monitor and design a cooling system Use motor sim software (e.g. Ansys Motor-CAD) to perform thermal analysis Add high-temperature automatic shutoff
Models are not accurate enough for good control	8	7	7	8	Measure metrics like linked rotor flux and compare to estimated flux to help debug; adjust for temperature, etc.
Electromagnetic noise interferes with sensor readings	6	9	9	6	Use proper shielding, decoupling, physically separate noisy and sensitive circuits Measure levels with EMI probe / RF field meter
Software loop cannot run fast enough	5	9	7	6.4	Look into different architectures, DMA
Shape of flux wave deviates from a sinusoid, leading to poorer performance	5	4	5	4	Use a winding jig to help distribute wires more uniformly or design PCB forcers

System Level Diagram



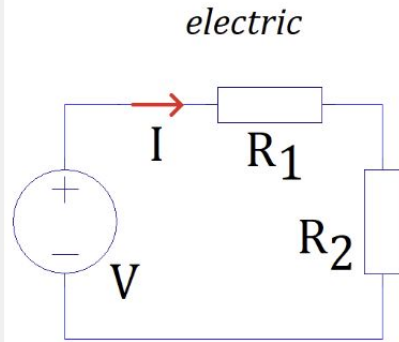
Any Questions?

Thank you for your support!

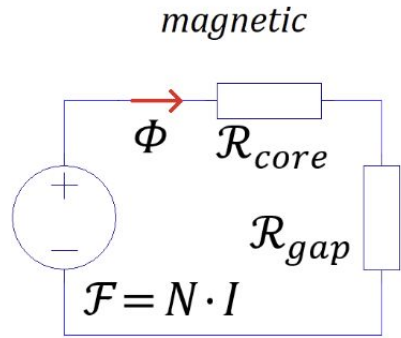
Appendix

Compare and contrasting Electric and Magnetic Circuits:

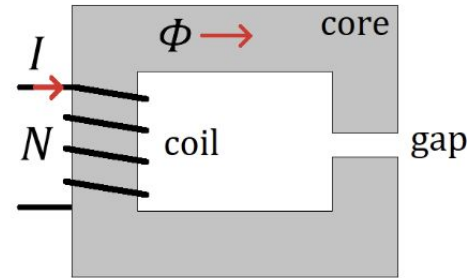
Simple electric circuit as an analogy of magnetic circuit



$$I = \frac{V}{R_1 + R_2}$$



$$\Phi = \frac{\mathcal{F}}{\mathcal{R}_{core} + \mathcal{R}_{gap}}$$



$$\mathcal{R} = \frac{l}{\mu_0 \mu_r A} = \frac{l}{\mu A}$$

Existing Designs II: Controller Setup for 3-phase FOC of induction motor

- ST Electronics provides [example hardware and firmware](#) for **Nucleo G431RB** (ARM Cortex-M4) board.
- Determine motor equivalent circuit from **DC**, **No-load**, and **Blocked Rotor** tests
- Create Simulink model to verify control system will work and tune PID constants.
 - Quickly test if the ST Electronics FOC example code works without modification.

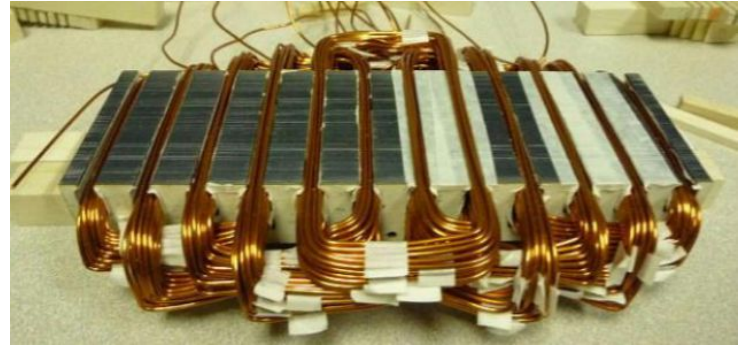


Parameter	Comment
R_s (Ω)	Stator phase winding resistance
L_{ls} (H)	Stator Leakage inductance
R_{Fe} (Ω)	Equivalent iron loss resistance
L_{ms} (H)	Magnetizing inductance
$L_{lr}^{\wedge}$ (H)	Rotor leakage inductance referred to stator
$R_r^{\wedge}$ (Ω)	Equivalent rotor phase resistance
Φ_n (Wb)	Nominal magnetizing flux

Existing Designs I: Prototype of induction linear motor and vector control drive circuit - Masaaki Kumagai

Takeaways:

- Starting point for stator geometry design
- How different parameters change performance of the motor
- High level example of a vector controller implementation



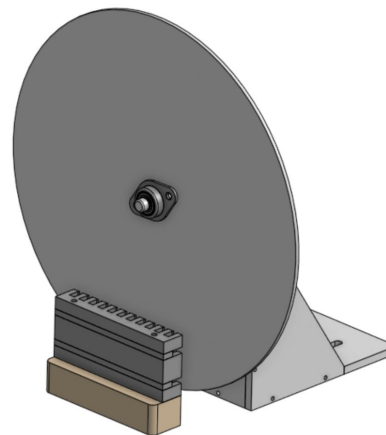
Static Test Setup

Prototype motor control on static test setup with linear stator and large cast iron wheel as rotor.

- Infinite running
- Don't need full test track and train

Goals:

- Test motor FOC speed control code (stable rpm / tangential speed)
- Verify our method of determining motor equivalent circuit
-



Rough Budget

Instinctive guess: ~\$400 for early test setup, \$300 for linear induction motor train, \$2000 for maglev

Add spreadsheet with tentative BOM for:

- Controller, power electronics & necessary sensors
 - (STM32G431RB board \$15 + Motor driver board \$241 + Expansion board \$10 = \$266 USD)
- Motor construction:
 - Enameled copper wire, tons of enameled copper wire! (Aluminium will probably do for early prototypes.) Wire length and gauge depend on desired magnetic flux & current loading.
 - Ferrite cores for stator windings. CAD up parts, make assembly plan.
- Any electrical, mechanical test equipment that Project Lab does not have
 - E.g. Dynamometer, current sensors / probes
- Machining of (Soft, unmagnetized) ferrite parts.
 - Google: Apparently a challenge due to brittleness! Reduce complicated machining work by buying cores in standard shapes, or reducing work to grinding and 'gapping' cores.

Dump

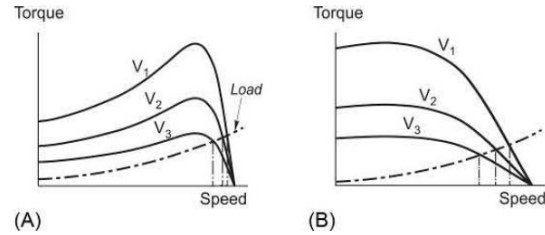


FIG. 6.21 Speed control of cage motor by stator voltage variation; (A) low resistance rotor, (B) high resistance rotor.

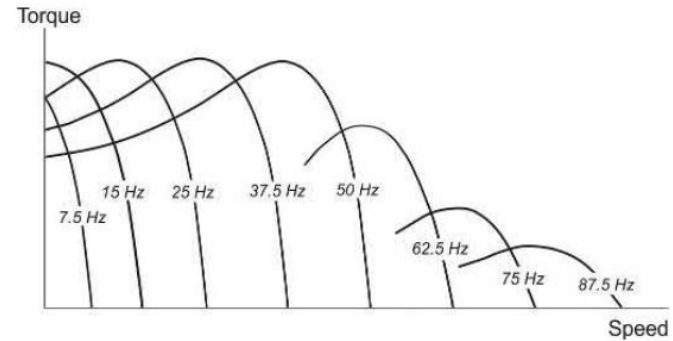


FIG. 7.4 Typical torque-speed curves for inverter-fed induction motor with constant flux up to base speed (50 Hz) and constant voltage at higher frequencies.

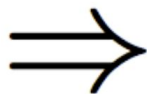
Fundamentals:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)$$



$$V = V_0 \cos(\omega t)$$

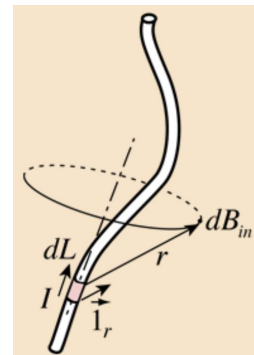
$$\Rightarrow I = \frac{V}{R}$$

$$\Rightarrow \vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\vec{s} \times \hat{r}}{r^2}$$



$$\varepsilon = \frac{d\Phi}{dt} \quad \Leftarrow$$

$$\Phi = \int \vec{B} \cdot d\vec{A}$$

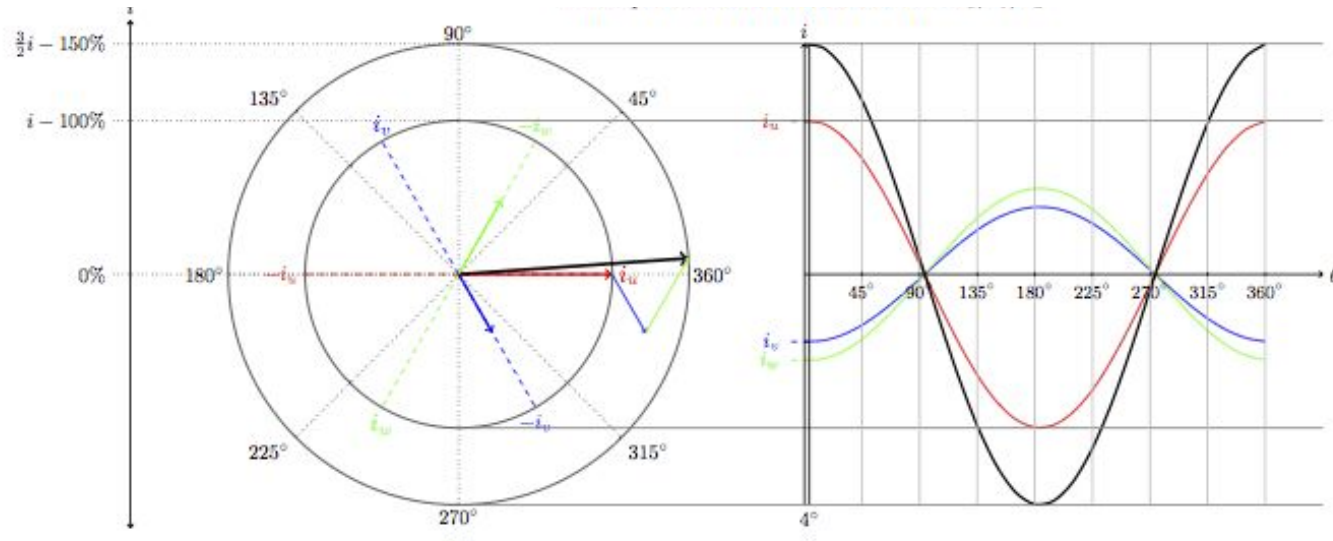


Deliverables and Scope Recap

Stage	Must Have Deliverables	Should Have Deliverables	Nice to Have Deliverables
Research	Conceptual summary of how induction motors work	Simulations to optimize design	
Initial build	A linear stator and a spinning rotor that can be driven by an off-the-shelf VFD	FOC controller (software and electronics) to more precisely control speed of rotor	Cooling system, UI for monitoring/debugging
Improvements	Summary of changes and their effect on benchmark test results		PCB forcer
Maglev (Stretch Goal)		Initial investigation into levitation and stability	Stator running along maglev track

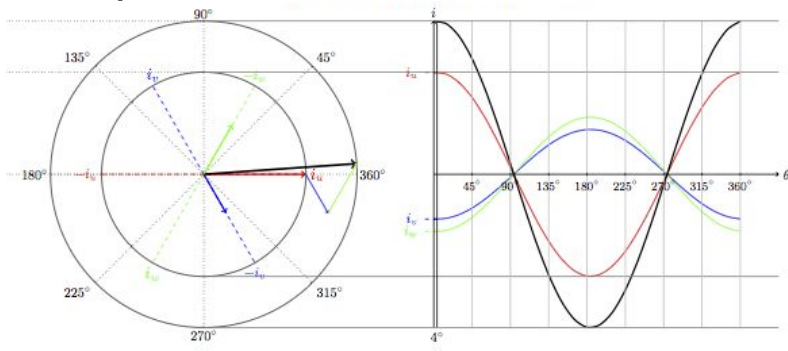
Overview of FOC - IV

- Therefore, we need to change the torque component of the stator current while keeping the flux component the same
- First, represent the travelling flux wave as a vector:



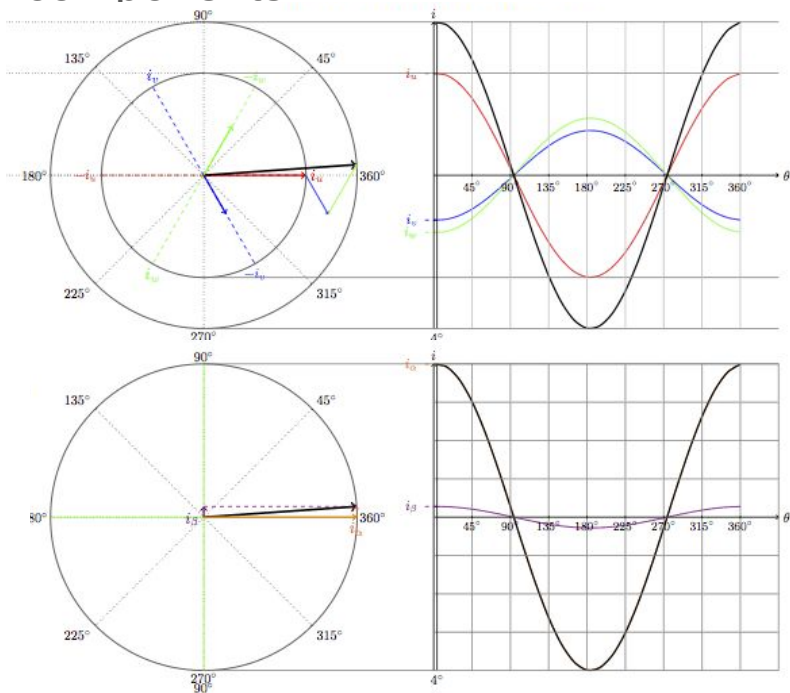
Overview of FOC - V

- Through the Clarke and Park transforms, we separate out the torque and flux components



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Overview of FOC - V

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